

Muhammad Shahid Jabbar

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[Current: Dhahran, Saudi Arabia. Permanent: Peoples Colony Multan]

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Summary

- Researcher with 8+ years of experience in Computer Vision, Medical Imaging, Inverse Problems, and Efficient Deep Learning, with a focus on resource-efficient and deployable AI models.
- Authored multiple SCIE-indexed publications in leading venues such as IEEE JBHI, Knowledge-Based Systems, and Information Processing & Management.
- Experience in teaching undergraduate courses, delivering AI/ML training programs, and co-supervising MS/PhD research.
- Active collaborator with international research groups, contributing to projects in multimodal AI, biomedical signal processing, and human-centered computing.

Work Experience

King Fahd University of Petroleum and Minerals

Post-Doctoral Fellow at SDAIA-KFUPM Joint Research Center for AI

Dhahran, Saudi Arabia

[04.2024] – [Present]

- Conducting research in Computer Vision and Medical Imaging, focusing on inverse problems and efficient deep learning, including ongoing work on diffusion-based models for image reconstruction, enhancement, and downstream vision tasks.
- Investigating resource-efficient and scalable methodologies for applying deep generative models in real-world and medical imaging applications.
- Co-supervising MS and PhD students in AI research in collaboration with faculty.
- Teaching undergraduate computing courses; delivered hands-on training in Data Science, Machine Learning, and Deep Learning, including workshops and intensive crash courses; mentored students through research projects, internships, and applied AI programs.
- Contributing to the organization of research activities and academic events within the SDAIA-KFUPM AI research center.

Radisen Co. Ltd.

Senior AI Engineer at AI Engineering Division

Seoul, South Korea

[10.2022] – [08.2023]

- Developed and optimized state-of-the-art AI-based digital radiography solutions for medical imaging applications.
- Developed weakly- and semi-supervised deep learning algorithms for triage of abnormalities and pathological findings.

Sungkyunkwan University

Researcher at H-Lab

Suwon, South Korea

[09.2016] – [08.2022]

- Developed LSTM-based pedestrian dead reckoning (PDR) for indoor positioning, achieving state-of-the-art 0.78 m mean error without external infrastructure.
- Designed sensor fusion and fingerprinting framework, improving localization accuracy by 87%, robust to signal degradation and environmental variations. Also, proposed BLE RSSI-based localization technique with 89% accuracy, robust to human and other obstacles.
- Proposed STFT+CNN-based islanding detection for multi-DG systems, achieving 98.76% accuracy with zero non-detection zone and robustness under noisy conditions (up to 30 dB).
- Developed multi-sensory reconfigurable assistive platform achieving 99.17% identification accuracy, validated through user studies (SUS, NASA-TLX).

Researcher at Media System Lab

[03.2021] – [08.2022]

- Developed fine-grained holistic textual-visual multi-modal representation learning method using foundation model transformers, with 47% improved mean retrieval ranking from CLIP zero-shot baseline.
- Developed neural attention models for automating the expert annotation and ranking task of multi-modal information retrieval.

Teaching Assistant at Sungkyunkwan University

[03.2019] – [06.2019]

- Courses: Electrical and Electronics Programming, Basics of Programming in C.
- Taught and supervised practical programming sessions in C language. Responsible for grading assignments and exams.
- Consistently received positive evaluations from students and professor.

Pakistan Telecommunication Company Ltd.

Engineer at Multimedia and Broadband Operations

Multan, Pakistan

[12.2013] – [08.2016]

- Worked on projects of Broadband services and Access Network optimization and rehabilitation.
- Conducted MSAGs, NGN, DSLAM and ONUs survey, development, operation and maintenance.
- Liaise with representatives of Huawei, ZTE and Alcatel for network expansion and the resolution of system-level issues.

Education

Sungkyunkwan University

Ph.D. in Engineering (Computer Engineering)

Suwon, South Korea

[09.2016] – [08.2022]

- CGPA: 4.14/4.5
- Relevant Coursework: Image Processing, Neural Networks, Machine Learning, Artificial Intelligence Based Simulation, Advanced Artificial Intelligence, HCI (Human Computer Interaction) Design, Estimation Theory, Advanced Computer Networks, Advanced Digital Communications.
- Thesis Title: Accessible and Scalable Visual-Art Appreciation Methods for Visually Impaired People.
- Program: MS leading to PhD program at Electrical and Computer Engineering Department.

- CGPA: 3.59/4
- Relevant Coursework: Neural Networks and Fuzzy Logic, Applications of Microcontrollers, Computer Communications Networks, Numerical Techniques, Digital Signal Processing and Filter Design, Microprocessor Interfacing Techniques, Probability and Statistics, Differential Equations and Partial Derivatives, Programming with C Language, Linear Algebra, Applied Calculus, Introduction to Computer.
- FYP Title: Disabled Patients Facilitation System.

Selected Funded Research Projects

Physics-Informed Data Preparation and Efficient Diffusion Modelling for High-Fidelity Medical Imaging

- Developing physics-guided diffusion architectures incorporating imaging operators (e.g., PSF, attenuation, noise models) for improved reconstruction fidelity and robustness.
- Designing computationally efficient diffusion models using few-step sampling, lightweight backbone, knowledge distillation, and quantization for real-time and edge deployment.
- Integrating generative outputs with downstream tasks (segmentation, classification, diagnostics) to enable task-aware performance improvements.

Spatio-Temporal Deep Learning for Robust and Efficient Face Presentation Attack Detection

- Developed spatio-temporal deep learning frameworks with squeeze-and-excitation and Bahdanau attention mechanisms for robust face presentation attack detection (Face Anti-Spoofing) under real-world conditions.
- Proposed multi-modal architectures augmenting RGB with synthetic depth modalities, achieving state-of-the-art performance with near-zero error rates on benchmark datasets.
- Designing lightweight models via knowledge distillation for deployment in resource-constrained and real-time systems.

Provoking LLMs by LLMs: A Red Team Framework for Proactive Detection of Harmful Content

- Investigated red teaming strategies for LLMs to proactively identify vulnerabilities and harmful behaviors.
- Analyzed attack taxonomies, benchmarks, and security risks across prompt-based and model-level adversarial techniques.
- Contributed to the development of evaluation frameworks for improving robustness, safety, and trustworthiness of generative AI systems.

Development of Non-Visual Information Transmission Interfaces for the Blind and Visually Impaired

- Contributed to the (successful) extension proposal for project, and the project report on completion.
- Developed a reconfigurable tactile platform as an assistive device for non-visual information transmission interface.
- Developed the deep learning-based AI-Ekphrasis framework which relies on transformer-based foundation models for retrieval of multi-modal image and fine-grained text data, exhibiting 8.2 of mean retrieval ranking as compared to the SOTA baseline model performance of 15.5.

Development of Robustness and Precision Improvement Technologies in Complex Indoor Location Estimation

- Contributed to the (successful) extension proposal for project the project report on completion.
- Responsible for IMU data measurement from test subjects, and development and testing of machine and deep learning-based solutions for robust indoor positioning and navigation algorithms.
- The developed LSTM based solution works for heading distance and heading direction estimation on, with a mean of 0.782m error in distance being most efficient at the time of development.

Selected Publications

Journal (SCIE)

- **Muhammad Shahid Jabbar**, Taha Hasan Masood Siddique, Kejie Huang, and Shujaat Khan. "Knowledge Distillation with Predicted Depth for Robust and Lightweight Face Presentation Attack Detection." Knowledge-Based Systems (2025): 114325.
- Adnan Munir, **Muhammad Shahid Jabbar**, and Shujaat Khan. "Daunet: A lightweight unet variant with deformable convolutions and parameter-free attention for medical image segmentation." IEEE Journal of Biomedical and Health Informatics (2026).
- **Muhammad Shahid Jabbar**, Sadam Al-Azani, Abrar Alotaibi, and Moataz Ahmed. "Red teaming large language models: A comprehensive review and critical analysis." Information Processing & Management 62.6 (2025): 104239.
- **Muhammad Shahid Jabbar**, Jitae Shin, Jun-Dong Cho. "AI Ekphrasis: Multi-Modal Learning with Foundation Models for Fine-Grained Poetry Retrieval." Electronics 11.8 (2022): 1275.
- **Muhammad Shahid Jabbar**, Chung-Heon Lee, Jun-Dong Cho. "ColorWatch: Color Perceptual Spatial Tactile Interface for People with Visual Impairments." Electronics 10.5 (2021): 596.
- Shin, Heejun, Taehee Kim, Juhyung Park, Hruthvik Raj, **Muhammad Shahid Jabbar**, et al. "Pulmonary abnormality screening on chest x-rays from different machine specifications: a generalized AI-based image manipulation pipeline." European radiology experimental 7.1 (2023): 68.
- Shin, Heejun, Taehee Kim, Hruthvik Raj, **Muhammad Shahid Jabbar**, et al. "Machine-independent AI for chest x-ray abnormality classification." Journal of Medical Imaging and Radiation Sciences 54.3 (2023): S12.
- Hussain, Arif, Chul-Hwan Kim, and **Muhammad Shahid Jabbar**. "An Intelligent Deep Convolutional Neural Networks-Based Islanding Detection for Multi-DG Systems." IEEE Access 10 (2022): 131920-131931.
- Hussain, Ghulam, Mukesh Kumar Maheshwari, Mudaser Latif Memon, **Muhammad Shahid Jabbar**, Kamran Javed. "A CNN based automated activity and food recognition using wearable sensor for preventive healthcare." Electronics 8.12 (2019).
- Ghulam Hussain, **Muhammad Shahid Jabbar**, Jun-Dong Cho, Sangmin Bae. "Indoor positioning system: A new approach based on LSTM and two stage activity classification." Electronics 8.4 (2019): 375.

International Conference (Presentations)

- Dongmyeong Lee, Hyegyeong Hwang, **Muhammad Shahid Jabbar**, Jun-Dong Cho. "Language of gleam: impressionism artwork automatic caption generation for people with visual impairments." Thirteenth International Conference on Machine Vision. Vol. 11605. SPIE, 2021. *(Best Paper Award)*
- **Muhammad Shahid Jabbar**, Ghulam Hussain, Jun-Dong Cho. "Indoor Positioning System: Improved deep learning approach based on LSTM and multi-stage activity classification." 2020 IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia). IEEE, 2020.
- Chung-Heon Lee, **Muhammad Shahid Jabbar**, and Jun-Dong Cho. "ColorWatch: Color-Tactile Interpretation to Improve Color Perception and Accessibility for People with Visual Impairments." 2020 International Conference on Information and Communication Technology Convergence (ICTC). IEEE, 2020.
- **Muhammad Shahid Jabbar**, Kamran Javed, Ghulam Hussain, Jun-Dong Cho, Seong Oun Hwang. "A novel deep learning approach on detection of fetal hypoxia." International Conference on Green and Human Information Technology (ICGHIT), 2020.
- Kamran Javed, Nizam Ud Din, **Muhammad Shahid Jabbar**, Ghulam Hussain, Seong Oun Hwang. "Shadow removal from facial images using generative adversarial network." International Conference on Green and Human Information Technology (ICGHIT), 2020.
- **Muhammad Shahid Jabbar**, Ghulam Hussain, Jun-Dong Cho, Sangmin Bae. "For Enhancing Robustness and Precision of WiFi Fingerprinting Indoor Positioning with Pedestrian Dead Reckoning." The 6th International Symposium on Advanced and Applied Convergence (ISAAC), 2018. *(Best Paper Award)*
- Ghulam Hussain, **Muhammad Shahid Jabbar**, Sangmin Bae, Jun Dong Cho. "Stress Detection of the Students Studying in University using Smartphone Sensors." The Fourth International Conference on Smart Portable, Wearable, Implantable and Disability-oriented Devices and Systems (SPWID), 2018.
- Ghulam Hussain, **Muhammad Shahid Jabbar**, Hyunwoo Kim, Jinwon Park, Jun-Dong Cho, Sangmin Bae. "Performance analysis of robustness in Bluetooth based indoor localization." The 3rd International Workshop on Convergence Information Technology (IWCIT), 2017. *(Best Paper Award)*

Patent

- Tactile device for displaying color information - 색상정보 표시용 촉각장치, Inventors: JD Cho, CH Lee, **MS Jabbar**. Patent office: KR, Patent number: 출원번호 10-2020-0148650 (접수번호 1-1-2020-1195342-23) (DAS 접근코드 8135).

Interests and Skills

Teaching Interests: (and related areas)

- Computer Vision
- Machine Learning
- Artificial Intelligence
- Deep Learning
- Natural Language Processing
- Computer Networks
- Digital Logic Design
- Advanced Digital Communications
- Object-Oriented Programming, Python, C/C++
- Data Science, Data Structures and Algorithms

Research Interests:

- Computer Vision and Medical Image Analysis
- Efficient Deep Learning (Lightweight Models, Knowledge Distillation, Diffusion Models)
- Inverse Problems and Image Reconstruction
- Multimodal and Transformer-based Learning

Technical Skills:

- Programming: Python, MATLAB, C++
- Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas
- Tools & Platforms: Linux, Git, Docker, Kubernetes

Languages:

- English (Full Professional Proficiency), Urdu (Native), Korean (Beginner), Arabic (Beginner)